

Lesson 17 — Fix-It — The Steps Are Jumbled

Matches LEGO Teacher Portal lesson: "Quack the Code".

Objective (I can...): I can find the step that's in the wrong place and fix the sequence so it works. **CS concept:** Debugging and developing a sequence (finding and fixing order mistakes) · **Builds on:** Lesson 16 (running a sequence).

AI Connection (2 min, read aloud): When AI acts weird, the people who build it hunt for the step that's out of order and fix it — just like us. *Ask the group:* "How could you tell WHICH step is the wrong one?"

Materials: LEGO CS & AI kit + device per pair, a printed "broken sequence" picture strip per pair. **Model reuse:** Sequences model from Lesson 16 — code only.

60-minute flow - Warm-up (10): "Silly order" — coach acts out a jumbled routine (shoes before socks) and kids shout the fix. Finding the mistake is called debugging. - **Concept + demo (8):** Show a short program whose steps are out of order so the job fails; think aloud, spot the wrong step, move it, and run again. (*Try "the silly robot."*) - **Build/Code (25):** Pairs get a "jumbled" program (a step in the wrong place), run it, see it fail, then **find and fix** the order so it works. Then they develop it: add one new step in the right spot. *Wiggle break + swap at ~12 min.* - **Challenge (7):** Make a sequence with a hidden order mistake and trade with another pair to fix each other's. - **Share + tidy-up (10):** Pairs show the "before" (broken) and "after" (fixed) and point to the step they moved. Model stays built.

Pair roles: Driver runs and fixes, Navigator watches for the wrong step and says where it goes. Swap at the wiggle break. (*Trio: third leads the hunt.*)

Differentiation: *Support* — only two steps are swapped. *Extension / fast-finisher* — fix a longer program with two mistakes. *Reteach* — run step by step and stop the second it looks wrong.

Exit check: Pair finds and fixes an order mistake so the sequence works, and can say what was wrong.

At-home (optional): Mix up the steps of a bedtime routine on purpose; have a grown-up fix the order. ~5 min.